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# A Thermal Stabilizer for $LiPF_6$-Based Electrolytes of Li-Ion Cells

**S. S. Zhang,*,z K. Xu,** and T. R. Jow***

*U.S. Army Research Laboratory, Adelphi, Maryland 20783, USA*

We report that tris(2,2,2-trifluoroethyl) phosphite (TTFP), in which the oxidization number of phosphor is three (III), can effectively improve thermal stability of $LiPF_6$-based electrolytes of Li-ion cells. Effect of TTFP on the performance of Li/graphite, Li/cathode, and Li-ion cells is estimated using cyclic voltammetry and galvanostatic cycling. Results show that TTFP is electrochemically stable in Li-ion chemistry. We also found that TTFP can prevent propylene carbonate (PC) decomposition and graphite exfoliation in Li/graphite half-cell. We propose a mechanism of TTFP protecting $LiPF_6$ and discuss the effect of TTFP on cell performance.


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Thermal instability of $LiPF_6$ has been recognized as one of the important origins for the high capacity fading and decreased calendar life of Li-ion cells at elevated temperatures ($>60°C$). It has been known that $LiPF_6$ thermally decomposes into gaseous $PF_5$ and LiF around 200°C.[1] The formed $PF_5$ is a very strong Lewis acid and it is highly reactive to many organic solvents. Furthermore, $PF_5$ can be readily hydrolyzed to produce HF, which has been identified as the source of cathode active material, especially spinel $LiMn_2O_4$, dissolution within Li-ion cells.[2] While in the presence of electrolyte solvents such as ethylene carbonate (EC) and dimethyl carbonate (DMC), the decomposing temperature of $LiPF_6$ can fall to as low as 85°C.[3] Such decompositions are characterized by the formation of solid precipitates (LiF), carbonate based polymers, and some gaseous products, which usually result in a brown solution. Therefore, many attempts have been made to replace thermally unstable $LiPF_6$ in the electrolytes of Li-ion cells. Among them, partial substitution of perfluoroalkyls for fluorides in $LiPF_6$ by electrochemical fluorination was reported to be very effective.[4] However, its synthesis is too expensive to be acceptable in the Li-ion industry. Other approaches include syntheses of new salts.[5-9] It appears that all these new salts must contain either phosphor (P)[5] or boron (B)[6-9] as a center atom because P and B are the most favorable for passivation of cathode current collector material (Al). Due mainly to the problem of Al corrosion, other salts such as lithium methides and lithium imides are not successful despite their high thermal stability. A new approach is to use the strong electron withdrawing anion ligands, which are boron-based compounds, to dissolve common lithium salts such as LiF by forming a 1:1 complex.[10,11] However, such ligands have rather high molecular weight and may have potential concerns with electrochemical stability in Li-ion chemistry.

We have found that tris(2,2,2-trifluoroethyl) phosphite (TTFP), which has an oxidization number of phosphor(III), can well protect $LiPF_6$ from being thermally decomposed in the electrolytes of Li-ion cells.[12] In this paper, we discuss the mechanism of TTFP protection and evaluate the electrochemical stability of TTFP in Li-ion cells.

## Experimental

$LiPF_6$ ($>99.9\%$, Stella Chemifa Corp.), battery grade ethylene carbonate (EC, Grant Chemical) and battery grade propylene carbonate (PC, Grant Chemical) were used as received. Ethylmethyl carbonate (EMC, water content $<30$ ppm, Mitsubishi Chemical Co.) and TTFP (Aldrich, 99%) were dried using activated alumina before use. Electrolytes with different compositions were prepared in an argon-filled glove box. Graphite and lithium nickel-based mixed oxide electrode films were supplied by SAFT America, Inc. and tris(pentafluorophenyl) borate (TPFPB) by Brookhaven National Laboratory. Both were dried under conditions recommended by the suppliers before use. Li/graphite, Li/cathode, and graphite/cathode cells with an electrode area of $6.0\ cm^2$ were assembled for electrochemical and cycling tests.

An EG&G PAR potentiostat/galvanostat model 273A was used for electrochemical measurements and a Maccor Series 4000 tester for galvanostatic cycling tests. The electrochemical window of the electrolytes was determined using a three-electrode cell consisting of Pt ($0.64\ cm^2$) working electrode, Li foil reference and counter electrodes, while other measurements were performed on two-electrode cells as assembled above. Testing conditions are shown in figure captions.

## Results and Discussion

*Improved stability of $LiPF_6$.*—Thermal stability of $LiPF_6$-based electrolytes is mainly affected by two factors, thermal instability of $LiPF_6$ and reactivity of the decomposed product ($PF_5$). There is evidence to show that the reactivity of $PF_5$ with electrolyte solvents plays a critical role in the thermal stability of Li-ion cell electrolytes. First, $PF_5$ is a very strong Lewis acid, which will catalyze degradation (decomposition or polymerization) of the electrolyte solvents. Second, it is also a strong fluorinating agent to react with organic solvents. Third, it can be readily hydrolyzed to form HF, which promotes chemical dissolution of the cathode active materials. Based on the above analyses, we believe that $PF_5$ is the main cause of thermal instability of $LiPF_6$-based electrolytes and that deactivation of $PF_5$ will increase the thermal stability. To confirm this hypothesis, we performed several experiments and observed the following phenomena.

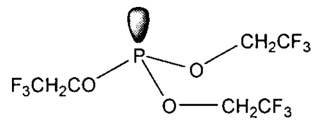
1. Presence of metal lithium significantly increased thermal stability of the electrolyte (1.2 M $LiPF_6$ 3:3:4 PC/EC/EMC, by weight). No color change was noticed after the electrolyte was stored at 60°C for 2 weeks. We ascribed this observation to the reaction of $2Li + PF_5 \rightarrow 2LiF + PF_3$, which considerably decreases the reactivity and Lewis acidity of $PF_5$.

2. Similar result was also obtained from lithium alkoxides, instead of metal lithium, because lithium alkoxides can deactivate $PF_5$ by reacting it to form LiF and more stable phosphates.

3. Addition of tris(pentafluorophenyl) borate (TPFPB) into the same electrolyte immediately changed the colorless electrolyte into a slightly brown solution. This color change is similar to that observed from the thermal degradation of the electrolyte. We believe that this is because TPFPB captures LiF from $LiPF_6$ to release $PF_5$ and the formed $PF_5$ subsequently reacts with electrolyte solvents (note that TPFPB is a very strong electron withdrawing ligand and it can form 1:1 complexes with many salts[10,11]).

* Electrochemical Society Active Member.
** Electrochemical Society Student Member.
z E-mail: szhang@mail.com

Based on above results, we used tris(2,2,2-trifluoroethyl) phosphite (TTFP) to deactivate $PF_5$. TTFP has the following structure




where the oxidization number of phosphor is three (III). As its structure shows, the phosphor center atom of TTFP carries a pair of lone-pair electrons, which are supposed to serve as electron donator to the complex with the strong electron withdrawing $PF_5$. Thus, the reactivity of $PF_5$ should be weakened by addition of TTFP into the electrolytes. Therefore, we prepared a series of 1.2 M $LiPF_6$ 3:3:4 PC/EC/EMC electrolytes without and with addition of TTFP and stored them in a 60°C oven. After a 2 week storage, we observed that the control electrolyte in the absence of TTFP had changed its color to brown while others containing 0.2, 0.5, 1.0, 2.0, and 5.0%, respectively, of TTFP remained colorless. These results indicate that deactivation of $PF_5$ indeed increases thermal stability of the $LiPF_6$-based electrolytes, which appears to be in conflict with the chemical reaction balance because deactivation (or removal) of $PF_5$ generally accelerates $LiPF_6$ decomposition ($LiPF_6 \rightarrow LiF + PF_5$). In our opinion, this is because thermal instability of the $LiPF_6$-based electrolytes mainly originates from the reactions between $PF_5$ and electrolyte solvents instead of the thermal decomposition of $LiPF_6$ salt. Although removal of $PF_5$ accelerates $LiPF_6$ decomposition, its effect on the thermal stability of the electrolyte solution may be too small to be reflected. Therefore, we conclude that TTFP can improve the thermal stability of the $LiPF_6$-based electrolytes.

*Electrochemical stability of TTFP*.—Electrochemical stability of TTFP is critical for its use in Li-ion cell electrolytes. Therefore, electrochemical stability of TTFP with respect to both oxidative and reductive potentials was evaluated using two fresh Pt electrodes, respectively. To increase solubility of $LiPF_6$ in the electrolyte, we used 1:1 PC/TTFP (wt) binary mixture as a solvent. Figure 1 shows that, at the oxidization side, the electrolyte can withstand up to 5.1 V, being higher than the potentials at which Li-ion cells normally operate. At the reduction side, electrochemical reduction of the electrolyte starts at 1.8 V. However, such a reduction is effectively suppressed with a further decrease in the potential, which is indicated by a wide current plateau between 1.8 V and the potential of metal lithium deposition at ~0 V *vs.* $Li^+$/Li. More interestingly, the suppressed reduction disappeared in the subsequent cycles. Therefore, we conclude that the above feature may be helpful for stabilization of PC-based electrolytes in Li-ion cells because TTFP is initially reduced at 1.8 V, being higher than the potentials (0.6-0.8 V) of PC normal decomposition on graphite surface. Therefore, it is possible that the suppressed reduction of TTFP helps graphite to form stable solid electrolyte interface (SEI) and then to prevent PC decomposition. Multicurrent peaks at ~0 V (Fig. 1) are ascribed to the formation and deformation of Li-Pt alloy. Such characteristics are often observed from Pt electrode with many Li ion conducting electrolytes.

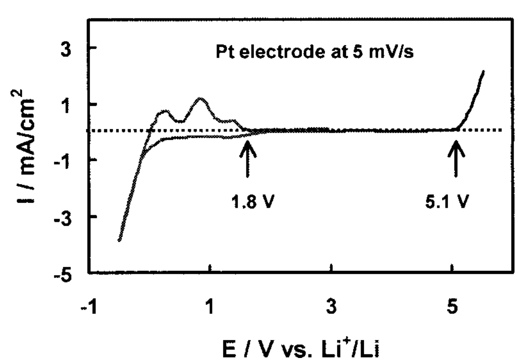



**Figure 1.** Electrochemical window of 1 M $LiPF_6$ 1:1 PC/TTFP electrolyte, determined using two fresh Pt electrodes at 5 mV/s.

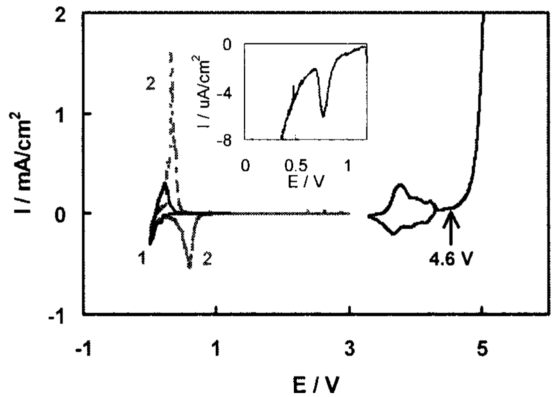



**Figure 2.** CVs of Li/graphite and Li/cathode half-cells using 1 M $LiPF_6$ 1:1 PC/TTFP electrolyte, recorded at 0.01 mV/s from newly assembled cells. For comparison, CV of Li/graphite half-cell with 1 M $LiPF_6$ 3:7 PC/EMC electrolyte is plotted as marked with number 2, and inset is the enlarged part of CV 1.

*Stability of TTFP against graphite and cathode*.—Electrochemical stability of TTFP is also examined using graphite and cathode, respectively, of Li-ion cells. We see from Fig. 2 that major oxidization of TTFP takes place at 4.6 V, which is slightly lower than that obtained from Pt electrode but still higher than the potentials of the cathode operation. It is shown that the oxidization current fell to the background before the solvent oxidization current starts increasing sharply at 4.6 V. On the other hand, we see a small reduction current peak from graphite electrode at 0.75 V (inset of Fig. 2). This reduction current may be contributed to the formation of SEI film on the graphite surface and the resultant SEI film prevents PC decomposition and graphite exfoliation. As a result, reversible intercalation and deintercalation of Li ions with graphite electrode take place in the voltage range of 0-0.5 V. To confirm the role of TTFP in helping the formation of SEI film, we measured cyclic voltammograms (CVs) of a Li/graphite half-cell using 1 M $LiPF_6$ 3:7 PC/EMC electrolyte (Fig. 2). There is no exception that PC is decomposed not only in the reduction process (~0.6 V) but also in the oxidization process (~0.3 V), as shown by two large current peaks.

Electrochemical stability of TTFP is further evaluated by cycling galvanostatically Li/graphite and Li/cathode half-cells, respectively. Figure 3 shows that, with 1 M $LiPF_6$ 1:1 PC/TTFP electrolyte, both the cathode and graphite can be well cycled in their operating voltage ranges, while the Li/graphite half-cell using 1 M $LiPF_6$ 3:7 PC/EMC electrolyte cannot be cycled. Voltages of this cell always remain above 0.5 V, as shown by curve 3, showing that PC molecules are continuously decomposed. In accordance with the CV results, these results indicate that TTFP protects PC molecules from being decomposed on graphite electrode.

*Performance of Li-ion cell*.—Figure 4 shows voltage-capacity curves of the initial two forming cycles for a graphite/cathode Li-ion cell using 1 M $LiPF_6$ 1:1 PC/TTFP electrolyte. Cycling efficiencies of the cell are evaluated to be 85 and 93% for the first and second cycle, respectively. In particular, cycling efficiency of the first cycles is slightly higher than that obtained from either Li/graphite or Li/cathode half-cell (Fig. 3). With an increase in the cycle number,

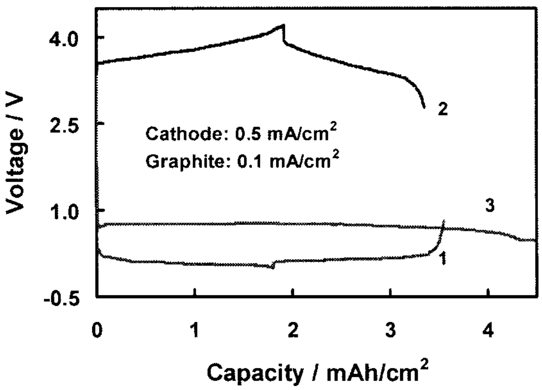



**Figure 3.** Voltage-capacity curves of Li/graphite and Li/cathode half-cells using 1 M $LiPF_6$ 1:1 PC/TTFP electrolyte, recorded from the first cycle. For comparison, the curve of Li/graphite half-cell with 1 M $LiPF_6$ 3:7 PC/EMC electrolyte is also plotted, as marked with number 3.

cycling efficiencies gradually increase and finally approach 100%. As can be seen in Fig. 4, there is a small amount of irreversible capacities in the initial period of the first charging process. These irreversible capacities can be better reflected in the plots of differential capacity *vs.* voltage (inset of Fig. 4), which shows that the irreversible capacities are mainly present below 3.7 V in the first charging process. The irreversible capacities observed from the first charging process of Li-ion cells can be more related to the decomposition of electrolyte solvents and to the formation of SEI film on the surfaces of both graphite[13] and cathode[14] electrodes. This could be why the cycling efficiency of the first cycle for Li-ion cells is often close or even slightly higher than that obtained from either Li/graphite or Li/cathode half-cell.

Figure 5 plots discharge capacities and cycling efficiencies of a graphite/cathode Li-ion cell as a function of the cycle number. Except for the initial few forming cycles, all following cycles show nearly unity of cycling efficiencies and the cell presents reasonable capacity retention with an increase in the cycle number. In addition to the benefits described above, we have noticed[12] that TTFP is an effective fire-retardant for the Li-ion cell electrolytes. When the amount of TTFP in the electrolytes reaches 15-20%, depending on

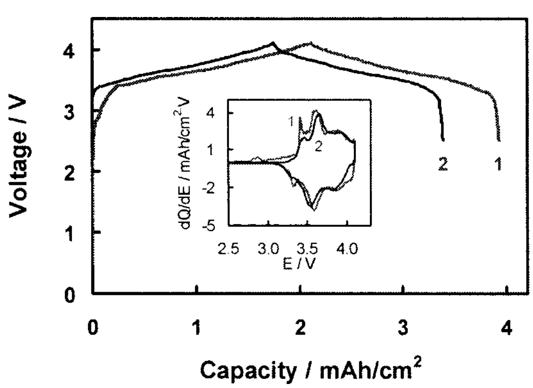



**Figure 4.** Voltage-capacity curves of graphite/cathode Li-ion cell with 1 M $LiPF_6$ 1:1 PC/TTFP electrolyte, recorded from the initial two forming cycles at 0.1 $mA/cm^2$. Inset shows differential capacity *vs.* cell voltage.

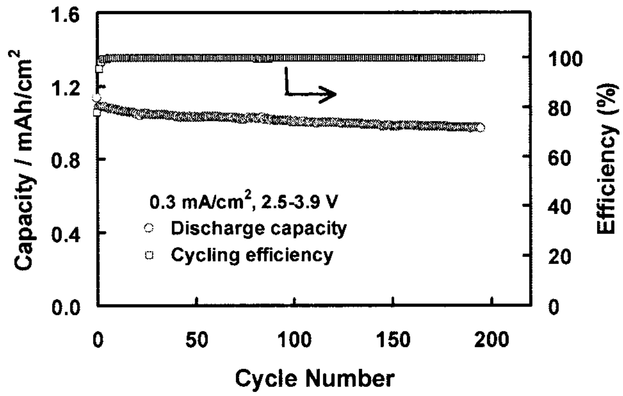



**Figure 5.** Discharge capacity and cycling efficiency of graphite/cathode Li-ion cell using 1 M $LiPF_6$ 1:1 PC/TTFP electrolyte *vs.* cycle number, obtained by cycling the cell at 0.3 $mA/cm^2$ between 2.5 and 3.9 V.

the electrolyte solvents used, the electrolytes become completely nonflammable. We will separately report the results about the effect of TTFP on fire-retardation and SEI formation and about the improvement of TTFP on cycling performance of the Li-ion cells at elevated temperatures ($>60°C$).

## Conclusions

Thermal instability of the $LiPF_6$-based electrolytes mainly originates from the reactions between the strong electron withdrawing $PF_5$ and electrolyte solvents. Thermal stability of the $LiPF_6$-based electrolytes can be effectively improved by adding appropriate amount, usually 1-5 wt % *vs.* the electrolyte solution, of tris(2,2,2-trifluoroethyl) phosphite (TTFP). In the voltage range of Li-ion cell operation, TTFP is electrochemically stable. Use of TTFP not only improves the thermal stability of the $LiPF_6$-based electrolytes but also facilitates the formation of solid electrolyte interface (SEI) on graphite surface.

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